

VISHING CHALLENGE · PRESENTATION ATTACK DETECTION

# Deepfake Voice Detection for Anti-Vishing Systems

A real-time, generalizable Presentation Attack Detection architecture  
that runs on the callee's phone and feeds a multi-layer phishing defense

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Solution Proposal · 2-Day Sprint · Panel Review



# Voice cloning has outpaced voice defense



## Cloning is now trivial

Modern TTS and voice-conversion models synthesize realistic speech from a few minutes of source audio, lowering the bar for impersonation attacks.



## Static detectors collapse on new generators

Detection methods trained on known synthesis methods fail to generalize across diverse, unseen generative models (Li & Chen, 2024).



## The stakes sit with the person answering

A cloned voice asking for an urgent transfer reaches the victim directly, with nothing behind it — a single miss is costly.

THE EVIDENCE

# How far it falls

TRAINED ON ASVSPOOF 2019, TESTED ON:

its own test set



another corpus

In-the-Wild



2026 cloned voices

over a real VoIP call



COIN FLIP



same test, retrained

on modern voices



equal error rate, lower is better



A 36-point gap between the benchmark and anything else. **Closing that gap is what we are building.**

## DIAGNOSIS

# It learns the dataset, not the voice

Before choosing an architecture we asked what the failing detectors are actually keying on. The answer explains the collapse, and it tells us what to remove during training.

### SILENCE ALONE

**85%**

accuracy from a classifier given only the length of the opening silence

Muller et al.

### ATTACK ON SILENCE ONLY

**82.2%**

attack success from changing only the background noise and the mute segments, never the voice

SiFDetectCracker, ACM MM 2023

### WHERE IT LOOKS

**0.5-0.6 kHz**

SHAP analysis: the model attends to the gaps between words and one narrow low band

arXiv:2110.03309

The detector wins on what belongs to the corpus, not to the speech. **Every one of those cues can be taken away in training.**

# What the panel is asking us to solve



## Objective

Develop and test a robust, generalized deep learning model for Presentation Attack Detection against modern audio deepfake and synthetic voice-spoofing techniques — improving voice authentication security while reducing multi-layer phishing risk.

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*Success = accuracy that holds up on generators it has never seen — not just the ones it trained on.*



## Real-time

Low enough latency for live-call integration



## Multi-layer ready

Feeds an adaptive phishing-defense stack



## 2-day sprint

Build, test, and prep an expert panel review



## 20-min readout

Findings + proposed solution to the panel

# The core bet: foundation models, not detectors from scratch

Systematic benchmarking across generators shows traditional detectors overfit to the synthesis methods they trained on — while large pretrained speech foundation models generalize far better, including across languages they were never fine-tuned on.

## Traditional CNN / RNN Detectors

- ✗ Trained end-to-end on spectrograms from known vocoders
- ✗ Strong on in-distribution benchmarks
- ✗ Accuracy drops sharply on unseen generator families
- ✗ Little transfer across languages or channel conditions

## SSL Foundation Models (Wav2Vec2 / WavLM)

- ✓ Large-scale pretraining gives richer, more general speech representations
- ✓ Demonstrated cross-generator generalization in benchmark studies
- ✓ Robust cross-lingual performance despite English-only fine-tuning
- ✓ Supports few-shot adaptation for high-value protected identities

# Detection pipeline, end to end



Decision layer: trust score is thresholded for real-time block/allow, and passed upstream as a signal into the multi-layer phishing-defense stack (see next).

# Making it real-time, not just accurate



## Streaming inference

Sliding 1–2s windows with overlap; running trust score updates instead of waiting for a full call.



## Model compression

Quantize/distill the SSL backbone (INT8, distilled student) so it keeps up with the live stream on a phone.



## Degraded-channel robustness

We measured the official baseline per codec: G.711 costs little (6.3%), but GSM hits 10.7% and uncontrolled PSTN transcoding 16.7%. Compression severity, not bandwidth, is what hurts.



## On-device triage

Fast handcrafted-feature triage on the device for an instant first pass; the heavier backbone confirms right after.

# One signal in a multi-layer phishing defense



## Layer 1 — Voice PAD

Real vs. synthetic score per call segment (this project's deliverable)



## Layer 2 — Behavioral & Context

Caller-ID spoofing signals, call metadata anomalies, urgency / funds-transfer request patterns typical of vishing scripts



## Layer 3 — Fusion & Adaptive Decision

Combines PAD score with contextual risk; borderline voice + high-risk context triggers step-up auth, high-confidence spoof triggers an immediate block

*Grounded in a multi-layer adaptive framework for phishing-attack detection (Ige, Kiekintveld & Piplai, 2024).*

# Proving generalization, not just accuracy



## EER, min t-DCF, per call-minute

Standard PAD metrics on held-out splits. And a streaming detector is judged per call, not per window: at 5% per-window error a 60-second call false-alarms 79% of the time.



## Leave-one-generator-out

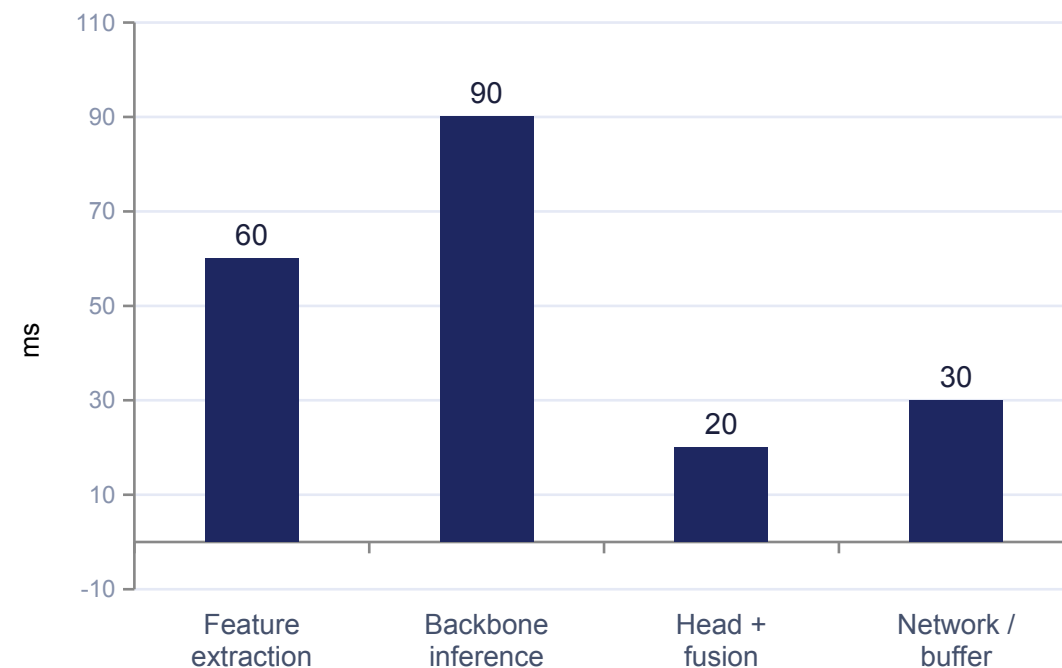
Entire vocoder / TTS families held out of training to directly test cross-generator generalization.



## Cross-lingual robustness (stretch)

Spot-check performance on non-English speech with no language-specific fine-tuning.

WHERE THE TIME GOES · MS PER 2-SECOND CHUNK



The alarm is read by a person, so the real budget is seconds. Per chunk we only have to keep up with the stream, which is what compression on slide 7 buys us.

# From proposal to a working detector

1

2-DAY SPRINT

## Literature & proposal

Survey the generalization failure, fix the evaluation protocol, and commit to an architecture. This deck.

2

WEEKS 2-4

## Baseline

Reproduce published clean-condition EER and validate against the official challenge baseline scores.

3

WEEKS 5-9

## Generalization

Degraded-channel and silence augmentation, foundation-model front-end, leave-one-generator-out. The core work.

4

WEEKS 10-12

## Real-time & write-up

Streaming inference, compression, latency on device, then the report and defence.

# Limitations & next steps



## Known limitations

- 2-day scope limits training data breadth vs. production-scale coverage
- Foundation-model backbones raise compute cost per stream vs. lightweight CNNs
- No dataset fully replicates live, adversarial vishing call conditions



## Next steps beyond the sprint

- ✓ Expand few-shot enrollment for high-value protected identities
- ✓ Field-test on real degraded-channel call recordings
- ✓ Formalize the fusion/decision policy with the phishing-defense team

# Questions for the panel

Deepfake Voice Detection for Anti-Vishing Systems · Solution Proposal

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Thank you.

